

1. A particle in the infinite square well has its initial wave function an even mixture of the first two stationary states:

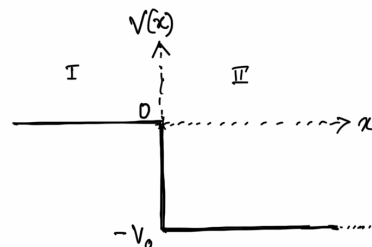
$$\psi(x, 0) = A[\psi_1(x) + \psi_2(x)].$$

- (a) Normalize  $\psi(x, 0)$ .
  - (b) Find  $\psi(x, t)$  and  $|\psi(x, t)|^2$ .
  - (c) Compute  $\langle x \rangle$ . Notice that it oscillates in time. What is the frequency of the oscillation? What is the amplitude of oscillation?
  - (d) Compute  $\langle p \rangle$ .
  - (e) Find the expectation value of  $H$ .
2. A potential barrier is defined by

$$V(x) = \begin{cases} 0 \text{ eV} & (\text{for } x < 0), \\ 7 \text{ eV} & (\text{for } x \geq 0). \end{cases}$$

A beam of electrons with energy 3 eV collides with this barrier from left. Find the value of  $x$  for which the probability of detecting the electron will be half the probability of detecting it at  $x = 0$ .

3. A particle of mass  $m$  and kinetic energy  $E > 0$  approaches an abrupt potential drop  $V_0$  (see figure).



- (a) What is the probability that it will *reflect* back, if  $E = V_0/3$ ?
  - (b) When a free neutron enters a nucleus, it experiences a sudden drop in potential energy, from  $V = 0$  outside to around -12 MeV inside. Suppose a neutron, emitted with kinetic energy 4 MeV by a fission event, strikes such a nucleus. What is the probability it will be absorbed (*i.e.* transmitted to the other side), thereby initiating another fission?
4. Imagine a bead (a quantum particle) of mass  $m$  that slides frictionlessly around a circular wire ring of circumference  $L$ . Find the stationary states (with appropriate normalisation) and corresponding allowed energies.  
[Hint: It is like a free particle having an additional constraint:  $\psi(x + L) = \psi(x)$ .]

5. We discuss in class about the commutator relation between the position ( $\hat{x}$ ) and momentum ( $\hat{p}_x$ ) operators as  $[\hat{x}, \hat{p}_x] = i\hbar$ . Show that in general (by method of induction),

$$[\hat{x}^n, \hat{p}_x] = ni\hbar\hat{x}^{n-1} \text{ and } [\hat{x}, \hat{p}_x^n] = ni\hbar\hat{p}_x^{n-1},$$

where  $n$  is a positive integer.

[Hints: Use the property of a commutator  $[AB, C] = [A, C]B + A[B, C].$ ]